Leo Lin

www.leo-lin.com | leo.daniel.lin@gmail.com | linkedin.com/in/leolin6/ | github.com/LeoLin6 | Cupertino, CA | 408-420-5768

EDUCATION

University of Illinois - Urbana Champaign — Aug. 2023 - Dec 2026
Bachelor of Science in Computer Engineering — GPA: 3.96/4.00
Awards: 1st place (x2) - NVIDIA/Lerobot hackathon; ECE Visionary Scholarship (top 8 out of 500 ECE majors)

WORK EXPERIENCE

Amazon Robotics @ Innovation Lab — Westborough, MA
Software Development Engineer Co-op — Jan 2026 - Jun 2026

- Built deterministic C++ control loops on PREEMPT_RT Linux, meeting Franka 1 kHz deadlines with sub-ms jitter
- Interfaced Manus gloves and Vision Pro to Isaac Sim via custom low-latency ROS 2 C++ bridges
- Synced real-time robot states to Isaac Sim digital twins using FoundationPose 6D tracking and SAM2
- Streamed 1,000+ multimodal teleop episodes into AWS S3 buckets, evaluating data pipelines to train and benchmark Interactive World Simulator vs. NVIDIA Cosmos under induced network latency

K-Scale Labs (YC Based Robotics startup) — Palo Alto, CA
Software Engineering/ML Intern — July 2025 - Aug 2025

- Developed Rust-based serial, UDP, and SocketCAN interfaces to stream teleop commands and actuator telemetry
- Wrote inference pipeline to control the robot real time in Mujoco Simulator using teleoperation setup
- Fine-tuned Gr00t-n1.5 VLA models and deployed end-to-end policies over CAN interfaces to execute language-conditioned cube-sorting demonstrations

Robnux — Sunnyvale, CA
Software Engineering Intern — May 2024 - Aug 2024

- Automated deployment across factory PCs w/ Docker-based synchronization between local & cloud environments

PUBLICATIONS

CRAFT: A Tendon-Driven Hand with Hybrid Hard-Soft Compliance — Robotics: Science and Systems (RSS) 2026

- Leo Lin (First Author), et al. Best Paper Award — RSS 2026 Workshop on Dexterous Manipulation
- Led experiments and evals for paper; designed tasks, collected data, & reported success metrics that drove paper claims
- Wrote firmware/API, FrankMocap+HaMeR vision teleop/retargeting for CRAFT and xArm

PolyFusion: collision-free robot learning with Bernstein polynomials — Amazon Robotics, CoRL 2026

- Co-authored a generative arm policy that predicts a short path from the scene and grasp pose, at ~80× faster runtime

RESEARCH

Mechatronics and Intelligent Robotics Lab | *NSF REU Research Fellow* — May 2025 - July 2025

- Combined YOLO object and hand detection with MoveIt/ROS for Sawyer bottle grasping and human handoff
- Streamed multi-modal tactile force and temperature telemetry over high-speed serial from an ESP32-S3 for closed-loop grasp detection

UIUC Augmented Listening Technology Lab | *Research Lead* | *Ki Cad* — Jan 2024 - May 2025

- Designed KiCad PCB board + Arduino/C firmware for a silent pan-tilt turret
- Reduced mechanical noise by ~40% via UART-tuned TMC motor drivers and custom designed PCB board

HACKATHON WINS

1st place - Sock Sorting Robot at LeRobot global hackathon | *Python, Pytorch, OpenCV, CUDA, AI agent* — Jun 2025

- Combined Gemini for high-level planning w/ diffusion models for low-level control to pair and sort socks.

Team Lead (1st place) - matcha making robot | NVIDIA x HF Hackathon | *Python, Huggingface* — Oct 2025

- Fine-tuned Gr00t N1.5 VLA model and contributed bimanual control pip packages to LeRobot library

TECHNICAL SKILLS

Python, C++, C, PyTorch, ROS/ROS2, Isaac Sim, MuJoCo, NumPy, Linux, Docker, Git, Arduino, UART, Fusion

Relevant Coursework: Computer Systems & Networking, Data Structures (C++), Control Sys (Simulink & MatLab), AI, Digital Systems Lab (FPGA), Probability & Stats, Object-Oriented (Python), Linear Algebra